Q1: What is the definition of an industrial robot according to the Robotics Industries Association (RIA)? A: An industrial robot is defined as a reprogrammable, multifunctional manipulator designed to move materials, parts, tools, or specialized devices through various programmed motions for the performance of a variety of tasks.

Q2: What is the difference between Direct (Forward) Kinematics and Inverse Kinematics? A: Direct Kinematics involves calculating the position and orientation of the robot's end-effector in the world coordinate system based on known joint angles and link parameters. Inverse Kinematics is the reverse process: determining the exact joint angles required to place the end-effector at a specific desired position and orientation in the workspace.

Q3: What are the Degrees of Freedom (DOF) in a robotic system? A: Degrees of Freedom refer to the number of independent parameters or joints that define the robot's configuration. In a typical robotic arm, each joint (whether rotary or prismatic) provides one degree of freedom. A minimum of 6 DOF is required to reach any point in a 3D space with any orientation.

Q4: What is a SCARA robot, and what are its primary applications? A: SCARA stands for Selective Compliance Assembly Robot Arm. It features two parallel rotary joints that provide compliance (flexibility) in the X-Y plane but rigidity in the Z-axis. SCARA robots are widely used for high-speed pick-and-place operations, packaging, and precise electronic assembly.

Q5: What is a Proportional-Integral-Derivative (PID) Controller? A: A PID controller is a control loop feedback mechanism used to maintain stable motion in robots. The Proportional (P) term reduces current error, the Integral (I) term accumulates past errors to eliminate steady-state drift, and the Derivative (D) term predicts future errors to dampen the system and prevent overshooting.

Q6: What is a Denavit-Hartenberg (DH) parameter matrix used for? A: DH parameters are a standard convention for attaching reference frames to the links of a spatial kinematic chain. By using four parameters (link length, link twist, joint distance, and joint angle), engineers can construct transformation matrices to solve the forward kinematics of any robotic arm mathematically.

Q7: What is an actuator in robotics, and what are the three main types? A: An actuator is a hardware device that converts stored energy into mechanical motion. The three main types used in industrial robotics are Electrical actuators (servo motors, stepper motors), Pneumatic actuators (using compressed air for fast, light loads), and Hydraulic actuators (using pressurized fluid for extremely heavy payloads).

Q8: What are the advantages and disadvantages of using a hydraulic drive system in a robot? A: Hydraulic drives provide a very high power-to-weight ratio, allowing the robot to lift massive payloads. They are also robust and self-lubricating. However, they are prone to fluid leaks, require high maintenance, generate noise, and cannot be used in clean-room environments.

Q9: How does a harmonic drive work, and why is it used in robotic joints? A: A harmonic drive is a specialized gear system consisting of a wave generator, a flexspline, and a circular spline. It is heavily used in robotic joints because it provides a high gear reduction ratio in a very compact space, features zero backlash (no play between gears), and offers high precision for positional control.

Q10: What is the function of a proximity sensor in automated manufacturing? A: A proximity sensor detects the presence or absence of an object without physical contact. They use electromagnetic fields, light, or sound to detect objects. They are used to prevent collisions, confirm that a part is correctly loaded into a machine, or trigger the next step in an assembly line.

Q11: How does an ultrasonic sensor measure distance? A: An ultrasonic sensor emits high-frequency sound waves (via a Trig pin) that bounce off an object and return to the sensor (via an Echo pin). By measuring the time it takes for the echo to return and knowing the speed of sound in air, the microcontroller can calculate the exact distance to the object.

Q12: What is an optical encoder, and what is the difference between absolute and incremental encoders? A: An optical encoder is a sensor attached to a motor shaft that tracks rotational position and speed. An incremental encoder generates a pulse for every degree of movement but loses its positional data if powered off. An absolute encoder outputs a unique digital code for every position, meaning it always knows its exact angle even after a power cycle.

Q13: What is tactile sensing, and where is it applied? A: Tactile sensing involves detecting physical contact, pressure, or texture. In robotics, tactile sensors are placed on the fingertips of a robotic gripper to provide force feedback, ensuring the robot holds fragile objects (like an egg or a glass tube) firmly enough not to drop them, but gently enough not to crush them.

Q14: Explain the concept of Robot Vision and "Frame Grabbers." A: Robot vision involves using cameras to capture images of the environment. A frame grabber is a hardware device that captures an individual digital still frame from a continuous analog or digital video feed. The robot's CPU then processes this frame using thresholding, edge detection, or segmentation to identify parts on a conveyor belt.

Q15: What is Simultaneous Localization and Mapping (SLAM)? A: SLAM is an algorithm that allows a mobile robot to build a map of an unknown environment while simultaneously keeping track of its current location within that map. It relies heavily on sensor fusion, combining data from LiDAR, cameras, and wheel encoders to navigate autonomously.

Q16: What is a Remote Center Compliance (RCC) device? A: An RCC device is a mechanical tool attached to a robot's wrist that provides lateral and rotational flexibility. It is primarily used in automated assembly (like inserting a tight-fitting peg into a hole) to compensate for minor mechanical misalignments and prevent the parts from jamming.

Q17: What is the difference between a microcontroller and a microprocessor in robotic control? A: A microprocessor acts as the central brain and requires external RAM, ROM, and I/O ports to function; it is used for complex computations like computer vision. A microcontroller has the CPU, memory, and I/O pins all integrated onto a single chip, making it ideal for executing low-level firmware tasks like reading sensors and driving motors.

Q18: What is Pulse Width Modulation (PWM) and how is it used with servo motors? A: PWM is a technique for controlling the average power delivered by an electrical signal by rapidly turning the switch on and off. For servo motors, a specific PWM pulse width (e.g., 1.5 milliseconds) tells the motor's internal circuitry exactly what angle to hold.

Q19: What is trajectory planning in robotic motion? A: Trajectory planning is the process of calculating the exact path, velocity, and acceleration a robot must take to move from point A to point B. It ensures the movement is smooth, avoids obstacles, and does not exceed the mechanical limits (torque and speed) of the motors.

Q20: What are the primary safety standards for industrial robots? A: Safety standards (such as ISO 10218) mandate strict design and operational protocols. This includes the installation of physical safety fences, light curtains, emergency stop (E-stop) buttons, and "Safe Torque Off" (STO) systems that instantly cut power to the motors if a human enters the robot's work envelope.